Chapter 1: Numerical Methods

In this chapter, we introduce a locally mass conserving numerical framework. A conforming mixed finite element method (MFEM) is proposed to solve the static Darcy-Stokes coupled methanics system. The proposed conforming MFEM captures the normal fluxes across element edges, ensuring compatibility with local mass conservation. The transport system is solved using a finite volume method (FVM), which inherently ensures local mass conservation.

We begin by introducing the mesh partition used for both the MFEM and the FVM schemes. Based on the mesh partition, two stable element pairs are presented, one for Darcy flow and one for Stokes flow. We then present a noval nonlinear WENO weighting scheme for the FVM. Error estimatea are provided for both the MFEM and the FVM schemes.

1.1 Quadrilateral Mesh

We consider a bounded computational domain $\Omega \subset \mathbb{R}^2$, where $\partial\Omega$ denotes the Lipschitz-continuous boundary of the domain. For a given $h > 0$, let $\mathcal{T}_h$ be a *triangulation* of the closure of the domain $\overline{\Omega}$ made of quadrilaterals E, i.e.,

$$\overline{\Omega} = \cup_{E \in \mathcal{T}_h} E, \quad h_E < h, \tag{1.1}$$

where two quadrilaterals are either disjoint or share one edge or one vertex. We assume that the quadrilateral mesh is uniformly shape regular. A formal definition of uniform shape regularity of a quadrilateral mesh is given in Girault and Raviart (2011).

Definition 1.1. For any quadrilateral $E \in \mathcal{T}_h$, form the triangle $T_i, \quad i \in \{0, 1, 2, 3\}$ by excluding vertex v_i from the quadrilateral element E, and use the remaining three vertices. Let

$$
\begin{aligned}
h_E &= \text{diameter of Element } E, \\
\rho_E &= 2 \min_{1 \le i \le 4} \{\text{diameter of largest circle inscribed in } T_i\}.
\end{aligned} \tag{1.2}
$$

A mesh partition $\mathcal{T}_h$ is considered to be uniformly shape regular if there exists a shape regularity parameter σ_* independent of $\mathcal{T}_h$, such that for all quadrilaterals $E \in \mathcal{T}_h$,

$$\rho_E / h_E \ge \sigma_* > 0. \tag{1.3}$$

Note that no subtriangle T_i of E has zero measure, i.e., each quadrilateral E is assumed to be strictly convex, not degenerating into a triangle, line or point.

The quadrilateral $E \in \mathcal{T}_h$ is referred to differently depending on the context. In the finite element method, a quadrilateral E is called an *element*, while in the finite volume method, it is typically referred to as a *cell*. For simplicity, we use the term *element* throughout this section.

In Figure 1.1, we show a reference element $\hat{E} = [-1, 1]^2$, a local reference element $\tilde{E}$ and a general quadrilateral element E. The edges of element E are denoted by e_i, $i \in \{0, 1, 2, 3\}$, and the vertices of the element E are denoted as v_i, $i \in \{0, 1, 2, 3\}$, both numbered in the counterclockwise order. We begin counting from zero, first to align with the indexing convention of C++, and second to facilitate representation in terms of modulo arithmetic, which also starts from zero. The vertices can be understood by $v_i = e_i \cap e_{i+1}$, where $i \equiv i \pmod 4$. Here, $\boldsymbol{\nu}_i$ and $\boldsymbol{\tau}_i$ are used to represent the unit normal and the unit tangent vector respectively. The unit normal vector $\boldsymbol{\nu}_i$ points outwards from the element E. The corresponding unit tangent $\boldsymbol{\tau}_i$ is obtained by rotating $\boldsymbol{\nu}_i$ counterclockwise, following the right-hand rule.

We assume the general quadrilateral element E can be obtained by scaling from the local reference element $\tilde{E}$ such that $\tilde{E} = E/H$, where H is the maximal edge length of E, i.e., the length of edge e_1 in Figure 1.1. Here, $|e_1| = H$ and $|\tilde{e}_1| = 1$. There exists a non-affine bijective and smooth map $\mathbf{F}_{\hat{E}} : \hat{E} \to \tilde{E}$ mapping the vertices of $\hat{E}$ to the vertices of $\tilde{E}$. Note that the unit normal and the unit tangent vectors are invariant under the scaling mapping H.

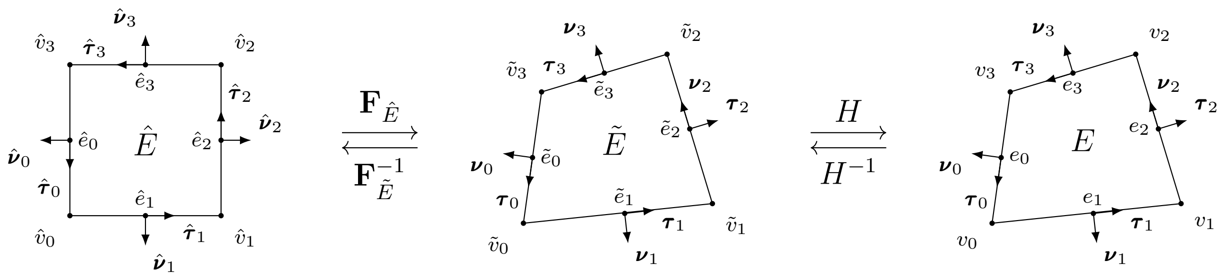


Figure 1.1: An exhibition of a reference square element, a local reference element, and a general non-degenerate quadrilateral element.

For the ease of implementation, we use a logically rectangular mesh, where each element is indexed using Cartesian coordinates $\{i, j\}$. In Figure 1.2, we illustrate a full logically rectangular, quasi-uniform quadrilateral mesh $\mathcal{T}_h$, which is generated by randomly distorting the interior vertices of a square mesh. The boundary vertices $v_i \in \partial\Omega$ remain fixed and are not subject to distortion.

1.2 Mathematical Preliminaries

In this section, we introduce some mathematical concepts that are relied upon in the remainder sections. The theory and notations are presented in a general dimension sense, but in practice, our discussion is confined to two dimension cases. In addition, for simplicity, we restrict our consideration to real-valued functions.

To begin with, we recall the definition of a Sobolev space.

Definition 1.2. Let $\Omega \subset \mathbb{R}^d$ be a domain, $1 \leq p \leq \infty$, and $m \geq 0$ be an integer. The Sobolev space of m derivatives in $L^p(\Omega)$ is

$$W_p^m(\Omega) = \{f \in L^p(\Omega) : D^\alpha f \in L^p(\Omega) \text{ for all multi-indices } \alpha \text{ such that } |\alpha| \leq m\}, \tag{1.4}$$

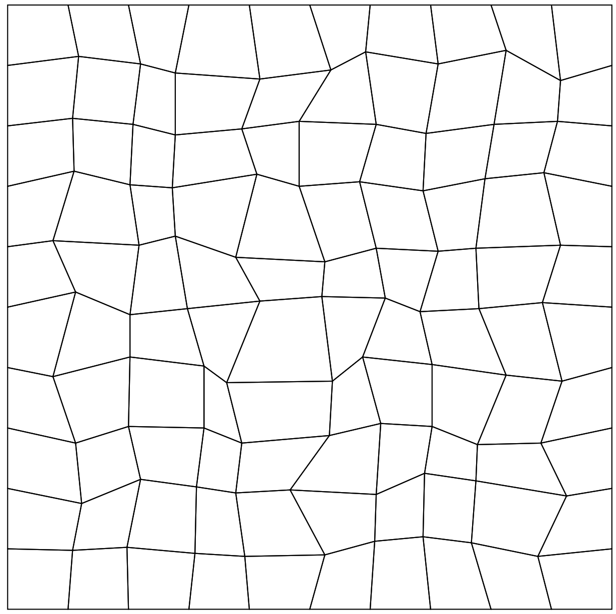


Figure 1.2: An exhibition of a full logically rectangular quasi-uniform quadrilateral mesh $\mathcal{T}_h$.

where the multi-index $\alpha = (\alpha_0, \alpha_1, \ldots, \alpha_{N-1}) \in \mathbb{N}^N$ and

$$|\alpha| = \sum_{i=0}^{N-1} \alpha_i.$$

The multi-index derivative is defined as

$$D^\alpha = \frac{\partial^{|\alpha|}}{\partial x_0^{\alpha_0} \ldots \partial x_{N-1}^{\alpha_{N-1}}}.$$

For $f \in L^p(\Omega)$, the $L^p(\Omega)$ norm is

$$\|f\|_{L^p(\Omega)} = \left(\int_\Omega |f|^p \right)^{1/p}. \tag{1.5}$$

We proceed with the definition of the Sobolev norm and seminorm.

Definition 1.3. For $f \in W_p^m(\Omega)$, the norm $\|\cdot\|_{W_p^m}$ is

$$\|f\|_{W_p^m(\Omega)} = \left\{ \sum_{|\alpha| \leq m} \|D^\alpha f\|_{L^p(\Omega)}^p \right\}^{1/p}, \quad p < \infty, \tag{1.6}$$

and

$$\|f\|_{W^m_\infty(\Omega)} = \max_{|\alpha|\le m} \|D^\alpha f\|_{L^\infty(\Omega)}, \quad p = \infty. \tag{1.7}$$

The seminorm $|\cdot|_{W^m_p}$ is

$$|f|_{W^m_p(\Omega)} = \left\{ \sum_{|\alpha|=m} \|D^\alpha f\|^p_{L^p(\Omega)} \right\}^{1/p}, \quad p < \infty. \tag{1.8}$$

In the special case when $p = 2$, we denote $W^m_2(\Omega) = H^m(\Omega)$. The corresponding norm and seminorn are then written as $\|\cdot\|_{H^m(\Omega)}$ and $|\cdot|_{H^m(\Omega)}$ respectively. Recall the Trace Theorem in $H^m(\mathbb{R}^d)$.

Theorem 1.1. *(Trace Theorem)*

Let k and d be integers with $0 < k < d$. The restriction map R extends to a bounded linear map from $H^m(\mathbb{R}^d)$ onto $H^{s-k/2}(\mathbb{R}^{d-k})$, provided that $s > k/2$.

Proof. See Arbogast and Bona (2025). □

Let $\mathbb{P}_r(\omega)$ denote the space of polynomials of degree up to r on $\omega \subset \mathbb{R}^d$. The dimension of $\mathbb{P}_r$ defined on $\mathbb{R}^d$ can be calculated as

$$\mathrm{dim}\mathbb{P}_r(\mathbb{R}^d) = \binom{r+d}{d} = \frac{(r+d)!}{r!d!}. \tag{1.9}$$

Let $\mathbb{Q}_{r,s}(\omega) :- \mathbb{P}_r(\omega) \otimes \mathbb{P}_s(\omega)$ denote the space of tensor product polynomials derived by polynomials of degree r and s on $\omega \subset \mathbb{R}^d$. The dimension of $\mathbb{Q}_{r,s}$ defined on $\mathbb{R}^d$ can be calculated as

$$\mathrm{dim}\mathbb{Q}_{r,s}(\mathbb{R}^d) = \mathrm{dim}\mathbb{P}_r(\mathbb{R}^d) \times \mathrm{dim}\mathbb{P}_s(\mathbb{R}^d). \tag{1.10}$$

We refer to the Bramble-Hilbert lemma for basic interpolation error estimation for polynomials of degree r.

Lemma 1.2. *(Bramble-Hilbert Lemma)*

Let $\Omega \in \mathbb{R}^d$ be a domain, and for function $f \in W_p^m(\Omega)$, $m \geq 0$, $1 \leq p < \infty$. There exists a constant C such that

$$\inf_{p\in\mathbb{P}_{k-1}(\Omega)} \|f - p\|_{W_p^m(\Omega)} \leq Ch^{k-m}|f|_{W_p^k(\Omega)}, \quad j = 0, 1, \ldots, k, \tag{1.11}$$

where the constant C has an upper bound due to the shape regularity assumption.

Proof. See Bramble and Hilbert (1970) and Dupont and Scott (1980). □

1.3 Direct Serendipity Space

Serendipity finite elements have the same accuracy of approximation with classical tensor product elements while using fewer degrees of freedom. However, the classical serendipity spaces suffer from poor approximation on quadrilaterals. To address this issue, we follow the approach of Arbogast et al. (2022) by defining local shape functions directly on quadrilateral elements, thereby recovering the desired accuracy. In the next sections, H^1 and $H(\text{div})$ conforming direct mixed finite element spaces are constructed with the direct serendipity function spaces.

To begin with, we introduce some auxiliary distance functions. We define the linear polynomial $\lambda_i(\mathbf{x})$ giving the distance of a point at coordinate $\mathbf{x} \in \mathbb{R}^2$ to edge e_i as

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_i^*) \cdot \boldsymbol{\nu}_i, \quad i = 0, 1, 2, 3, \tag{1.12}$$

where $\mathbf{x}_i^* \in e_i$ is any point on edge e_i. We pick $\mathbf{x}_i^* = \mathbf{x}_{v,i}$ for simplicity, where $\mathbf{x}_{v,i}$ denotes the coordinates of corner v_i. The distance function λ_i returns a positive value as long as the point is located in the interior of the element E.

We define two additional distance functions giving the distance of a point at coordinate $\mathbf{x} \in \mathbb{R}^2$ to the diagonal line d_i joining v_i with v_{i+2}, $i \in 0, 1$. Let $\boldsymbol{\nu}_{d,i}$ be the unit normal vector to the diagonal line d_i. The direction of the $\nu_{d,i}$ is defined by rotating counterclockwise of the unit vector pointing from v_i to v_{i+2}. The resulting diagonal distance function is defined as

$$\lambda_{d,i}(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}^*_{d,i}) \cdot \boldsymbol{\nu}_{d,i}, \quad i = 0, 1, \tag{1.13}$$

where $\mathbf{x}^*_{d,i}$ is any point on the diagonal line d_i. Here, we use the corner point v_i for simplicity, i.e., $\mathbf{x}^*_{d,i} = \mathbf{x}_{v,i}$.

We now present a set of nodal basis functions for the first and second order direct serendipity spaces, $\mathcal{DS}_1$ and $\mathcal{DS}_2$.

1.3.1 Second-order Space $\mathcal{DS}_2$

The dimension of the second-order polynomials in two dimension is calculated by (1.9) as

$$\dim\mathbb{P}_2(\mathbb{R}^2) = \binom{2+2}{2} = \frac{(4)!}{2!2!} = 6, \tag{1.14}$$

while a quadrilateral element requires a minimum of 4 vertex dofs and 4 edge dofs. Therefore, we supplement polynomial space $\mathbb{P}_2(E)$ with two linearly independent functions, $\phi_{s,1}(\mathbf{x})$ and $\phi_{s,2}(\mathbf{x})$, spanning the supplemental function space $\mathbb{S}^{\mathcal{DS}}_r(E) = \text{span}\{\phi_{s,1}, \phi_{s,2}\}, \ r \geq 2$. We present the second-order direct serendipity space as

$$\mathcal{DS}_2(E) = \mathbb{P}_2(E) \oplus \mathbb{S}^{\mathcal{DS}}_2(E). \tag{1.15}$$

The supplemental functions are defined as

$$\phi_{s,i}(\mathbf{x}) = \lambda_{i+1}(\mathbf{x})\lambda_{i+2}(\mathbf{x})R_{i,i+2}(\mathbf{x}), \quad i \in \{0, 1\}, \tag{1.16}$$

where $i \equiv i \pmod 4$. We use a simple definition for rational function $R_{i,i+2}$ as

$$R_{i,i+2}(\mathbf{x}) = \frac{\lambda_i(\mathbf{x}) - \lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \tag{1.17}$$

which satisfies the conditions

$$R_{i,i+2}|_{e_i} = -1, \quad R_{i,i+2}|_{e_{i+2}} = 1. \tag{1.18}$$

We continue to define

$$R_i(\mathbf{x}) = \frac{1}{2}\Big(1 - R_{i,i+2}(\mathbf{x})\Big) = \frac{\lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \quad i \in \{0,1,2,3\}, \tag{1.19}$$

where $i \equiv i \pmod 4$ as usual, and $R_{i,i+2} = -R_{i+2,i}$ according to the equation (1.17). Here, $R_i|_{e_i} = 1$, and $R_i|_{e_{i+2}} = 0$ by definition. In Figure 1.3, we draw R_0 on a general quadrilateral element, which evaluates to 1 on edge e_0, and evaluates to 0 on the opposite edge e_2.

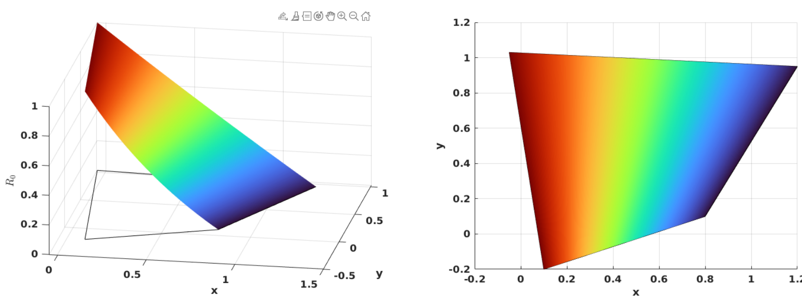


Figure 1.3: Rational function R_0.

The edge nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi^{(2)}_{e,i}(\mathbf{x}) = \frac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})R_i(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.20}$$

where $\mathbf{x}_{e,i}$ are the coordinates of the middle point of the edge e_i, again $i \equiv i \pmod 4$. We use superscript (2) to denote the order of the function space. We can restrict the edge basis function to edges as

$$\varphi^{(2)}_{e,i}|_{e,j} = \begin{cases} \dfrac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})}, & i = j \\ 0, & i \neq j \end{cases} \tag{1.21}$$

In Figure 1.4, we present edge basis function $\varphi^{(2)}_{e,0}$ on a general quadrilateral element, where $\varphi^{(2)}_{e_0}$ evaluates 0 on edges $\{e_i, \quad i \neq 0\}$, and evaluates to a quadratic function on edge e_0.

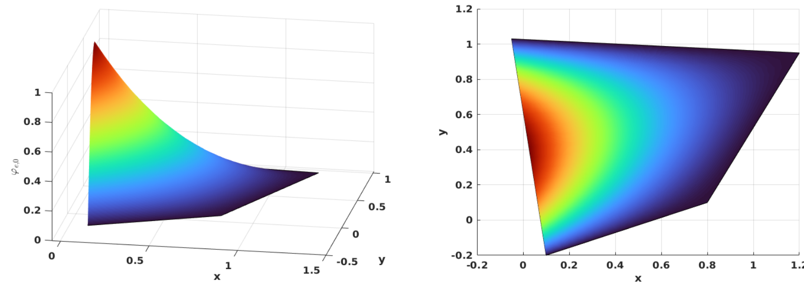


Figure 1.4: Edge nodal basis function $\varphi_{e,0}^{(2)}$ in $\mathcal{DS}_2$.

The vertex nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{v,i}^{(2)}(\mathbf{x}) = \frac{\lambda_{i+2}(\mathbf{x})\lambda_{i+3}(\mathbf{x}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}^{2}(\mathbf{x})}{\lambda_{i+2}(\mathbf{x}_{v,i})\lambda_{i+3}(\mathbf{x}_{v,i}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}^{(2)}(\mathbf{x}_{v,i})}, \tag{1.22}$$

where $\mathbf{x}_{v,i}$ is the coordinates of the vertex node v_i, and $i \equiv i \pmod 4$ as well. In Figure 1.5, we show $\varphi_{v,0}^{(2)}$ on a general quadrilateral.

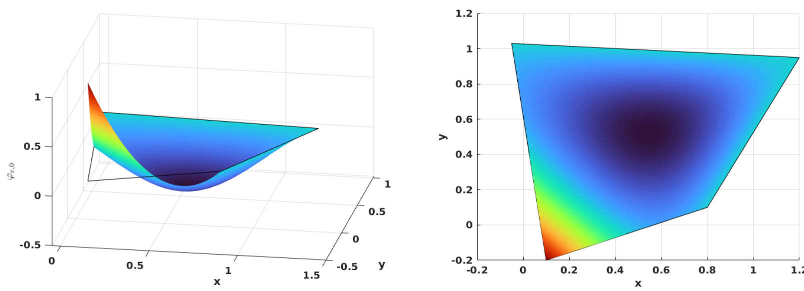


Figure 1.5: Vertex nodal basis function $\varphi_{v,0}^{2}$ in $\mathcal{DS}_2$.

We conclude that unisolvence follows directly from the constructed nodal basis functions.

1.3.2 First-order Space $\mathcal{DS}_1$

The dimension of the first-order polynomials in two dimension is also calculated by equation 1.9 as

$$\dim \mathbb{P}_1(\mathbb{R}^2) = \binom{1+2}{2} = \frac{(3)!}{2!1!} = 3, \tag{1.23}$$

while a quadrilateral element requires a minimum of 4 vertex dofs. Therefore, we supplement polynomial space $\mathbb{P}_1(E)$ with one additional function. The supplemental function for $\mathcal{DS}_1$ differs from that for $\mathcal{DS}_2$. We present the first order direct serendipity space as

$$\mathcal{DS}_1(E) = \mathbb{P}_1(E) \oplus \operatorname{span}\{R\}, \tag{1.24}$$

where $R(\mathbf{x}_{v,0}) = R(\mathbf{x}_{v,2}) = -R(\mathbf{x}_{v,1}) = -R(\mathbf{x}_{v,3}) = 1$. The construction of the rational function R is not unique. We present one way to construct $R(\mathbf{x})$ using edge nodal basis function (1.20) in $\mathcal{DS}_2$ as

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=0}^{3} r(\mathbf{x}_{e,i})\varphi_{e,i}^{(2)}(\mathbf{x}), \tag{1.25}$$

where the auxiliary function $r(\mathbf{x})$ is defined as

$$r(\mathbf{x}) = \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,0})\lambda_3(\mathbf{x}_{v,0})} - \frac{\lambda_0(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,1})\lambda_3(\mathbf{x}_{v,1})} + \frac{\lambda_0(\mathbf{x})\lambda_1(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,2})\lambda_1(\mathbf{x}_{v,2})} - \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})}. \tag{1.26}$$

In Figure 1.6, we plot the rational function $R(\mathbf{x})$ on a general quadrilateral.

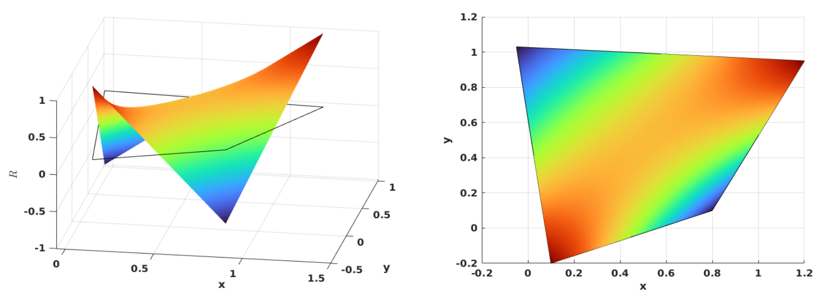


Figure 1.6: Supplemental rational function R in $\mathcal{DS}_1$.

We then define vertex nodal basis function accordingly as

$$\varphi_{v,i}^{(1)}(\mathbf{x}) = \frac{\lambda_{d,m}(\mathbf{x}) - \frac{1}{2}\lambda_{d,m}(\mathbf{x}_{v,i+2})(1 + (-1)^i R(\mathbf{x}))}{\lambda_{d,m}(\mathbf{x}_{v,i}) - \lambda_{d,m}(\mathbf{x}_{v,i+2})}, \tag{1.27}$$

where $m \equiv i+1 \pmod 2$, and $i \equiv i \pmod 4$. In Figure 1.7, we show the vertex nodal basis function $\varphi_{v,0}^{(1)}$ in $\mathcal{DS}_1$ on a general quadrilateral, which evaluates 1 at vertex v_0 and 0 at other vertices.

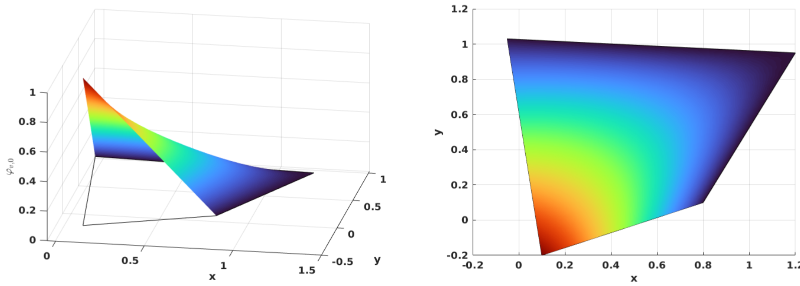


Figure 1.7: Vertex nodal basis function in $\mathcal{DS}_1$.

1.3.3 An Interpolation Operator

For completeness, we describe an interpolation operator mapping onto $\mathcal{DS}_r(K_i)$, $r = 1, 2$ following Arbogast et al. (2022). For each node a_i, the associated K_i has different meanings depending on the type of the nodal basis functions. If the basis function is associated with an edge node, we set $K_i = e_i$, where e_i is the corresponding edge. For the basis function associated with a vertex node, we set K_i to be any edge e containing node a_i, with the restriction that if $a_i \in \partial\Omega$, then $e \in \partial\Omega$.

An interpolation operator $\mathcal{I}_h^r : W_p^l(\Omega) \to \mathcal{DS}_r$ is defined as

$$\mathcal{I}_h^r v(\mathbf{x}) = \sum_{i=1}^{N_r} \varphi_i(\mathbf{x}) \int_{K_i} \psi_i(\mathbf{y}) v(\mathbf{y})\, d\mathbf{y} \in \mathcal{DS}_r, \tag{1.28}$$

where $1 \le p \le \infty$ and $l > 1/p$ (but $l \ge 1$ if $p = 1$), and $\psi(\mathbf{x})$ is the dual nodal basis such that

$$\int_{K_i} \psi_i(\mathbf{x}) \varphi_j(\mathbf{x})\, d\mathbf{x} = \delta_{i,j}, \quad i, j \in \{0, 1, 2, \ldots, N_r\}, \tag{1.29}$$

where N_r is the number of nodal basis functions. We present the following two lemmas regarding the constructed interpolation operator.

Lemma 1.3. *(Boundedness of the interpolation operator.)*

Let $\mathcal{T}_h$ be uniformly shape regular according to Definition 1.3 with shape regularity parameter σ_*. Let $v \in W_p^l(\Omega)$, with $1 \le p \le \infty$ and $l > q/p$. Suppose that the basis functions of $\mathcal{DS}_r(E)$ are constructed using λ_{24} and λ_{13} such that the zero *set $\mathcal{Z}_{i,i+2}$ intersects e_{i+3} and e_{i+1}. Moreover, assume that $R_{i,i+2}$ is a uniformly differentiable functions of the vertices of E up to order m. Then for $r \geq 1$, $E \in \mathcal{T}_h$, $1 \leq q \leq \infty$, and any non negative integer m,*

$$\|\mathcal{I}_h^r v\|_{W_q^m(E)} \leq C(\sigma_*, m, q) \sum_{k=0}^{l} h_E^{k-m+\frac{2}{q}-\frac{2}{p}} |v|_{W_p^k(E^*)}, \tag{1.30}$$

where $E^* = \cup_{F\in\mathcal{T}_h, F\cap E\neq\emptyset} F$ and $|\cdot|_{W_p^k}$ is the seminorm of k-th order derivatives.

Proof. See Arbogast et al. (2022). □

Combining Lemma 1.3 and Bramble-Hilbert lemma 1.2, we obtain local and global error estimation.

Lemma 1.4. *(Approximability of the interpolation operator.)*

Under the assumption of Lemma 1.3, there exists a constant $C = C(r, \sigma_*) > 0$ such that for all functions $v \in W_{l,p}(E^*)$, with $1 \leq p \leq \infty$ and $l > 1/p$ or ($l \geq 1$ if $p = 1$),

$$\|v - \mathcal{I}_h^r v\|_{W_p^m(E)} \leq C h_E^{l-m} |v|_{W_p^l(E^*)}, \quad 0 \leq m \leq \min(l, r+1). \tag{1.31}$$

Moreover, there exists a constant $C = C(r, \sigma_*) > 0$, independent of $h = \max_{E\in\mathcal{T}}$, such that for all functions $v \in W_p^l(\Omega)$,

$$\left(\sum_{E\in\mathcal{T}_h} \|v - \mathcal{I}_h^r v\|_{W_p^m(E)}^p \right)^{1/p} \leq C h^{l-m} |v|_{W_p^l(\Omega)}, \quad 0 \leq m \leq \min(l, r+1). \tag{1.32}$$

Proof. See Arbogast et al. (2022). □

1.4 Mixed Finite Elements

In this section, we prepare mixed finite element space for the scaled coupled system (**??**) to (**??**). We begin by presenting the mixed variational formulations of the Stokes and Darcy model problems separately. Subsequently, we introduce H^1- and $H(\mathrm{div})$-conforming mixed finite element spaces.

We then describe the construction of mixed finite elements from the introduced direct serendipity space. Recall Ciarlet's definition Ciarlet (2002) of a finite element.

Definition 1.4. A triple $(E, X(E), \varphi_j)$ defines a finite element.

- Element $E \in \mathcal{T}_h$;
- Space of shape functions $X(E)$, and a set of shape function $\varphi_i \in X(E)$. The dimension of the shape function space $\dim(X(E)) = n$;
- A set of continuous and linear functionals ϕ_j defined on a subset $\mathcal{X}(E)$ of the energy space containing the finite element space $X(E)$ as

$$\phi_j : \mathcal{X}(E) \to \mathbb{R}, \tag{1.33}$$

such that unisolvence condition is achieved as

$$\langle \phi_j, \varphi_i \rangle = \delta_{ij}, \quad i, j \in \{1, 2, 3, \ldots, n\} \tag{1.34}$$

In latter subsections, we give explicit construction of nodal basis functions and degrees of freedom that is associated to them, thereby completing the definition of the finite element.

1.4.1 A Darcy Model Problem

Consider a model problem for Darcy flow equipped with homogeneous boundary condition as

$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{f}, \text{ in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0, \text{ in } \Omega, \\ \mathbf{u} &= \mathbf{0}, \text{ on } \partial\Omega. \end{aligned} \tag{1.35}$$

We seek displacement pressure solution pair $(\mathbf{u}, p)$ in identical test and trial spaces as

$$\begin{aligned} \mathbb{V}_D &= \{\mathbf{v} \in (H(\mathrm{div};\Omega))^2 : \mathbf{v} \cdot \boldsymbol{\nu} = \mathbf{0} \text{ on } \partial\Omega\}, \\ \mathbb{U}_D &= \{\mathbf{u} \in (H(\mathrm{div};\Omega))^2 : \mathbf{u} \cdot \boldsymbol{\nu} = \mathbf{0} \text{ on } \partial\Omega\} = \mathbb{V}_D, \\ \mathbb{W}_D &= \{p \in L^2(\Omega) : \int_\Omega p d\mathbf{x} = 0\} = L^2(\Omega)/\mathbb{R}. \end{aligned} \tag{1.36}$$

We write weak form with the defined energy spaces. The standard mixed weak form reads as follows:

Find $\mathbf{u} \in \mathbb{U}_D$ and $p \in \mathbb{W}_D$ such that

$$\begin{aligned} (\mathbf{u}, \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}), \quad & \forall \mathbf{v} \in \mathbb{V}_D, \\ (\nabla \cdot \mathbf{u}, \omega) = 0, \quad & \forall \omega \in \mathbb{W}_D. \end{aligned} \tag{1.37}$$

We introduce two finite-dimensional subspace $\mathbb{V}_{D,h} \subset \mathbb{V}_D$ and $\mathbb{W}_{D,h} \subset \mathbb{W}_D$, and consider the discretized problem:

Find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ such that

$$\begin{aligned} (\mathbf{u}_h, \mathbf{v}_h) - (p_h, \nabla \cdot \mathbf{v}_h) = (\mathbf{f}, \mathbf{v}_h), \quad & \forall \mathbf{v}_h \in \mathbb{V}_{D,h}, \\ (\nabla \cdot \mathbf{u}_h, \omega_h) = 0, \quad & \forall \omega_h \in \mathbb{W}_{D,h}. \end{aligned} \tag{1.38}$$

Now we present a construction of discretized space pair $\mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ for this Darcy model problem in the next subsection.

1.4.2 A reduced $H(\mathbf{div})$ conforming space $\mathbb{V}_1^0$

We have some choice to discretize the space $\mathbb{V}_D$. For example, we can choose the classical Ravier-Thomas element Raviart and Thomas (1977). The serendipity space is the natural precursor of the Brezzi-Douglas-Marini element Brezzi et al. (1985) related by a de Rham complex. We generated a reduced first-order $H(\text{div})$ approximation from the direct serendipity space, that $\nabla \cdot \mathbf{u}$ is approximated one less order than approximation of $\mathbf{u}$. We can also recover Arbogast-Correa element Arbogast and Correa (2016) if we use the mapped serendipity space.

We write the rotated 2D exact sequence for related energy space as

$$\begin{array}{ccccc} H^1 & \xrightarrow{\text{curl}} & H(\text{div}) & \xrightarrow{\text{div}} & L^2 \\ \cup & & \cup & & \cup \\ \mathcal{DS}_2 & \xrightarrow{\text{curl}} & \mathbf{V}_r^{r-1} & \xrightarrow{\text{div}} & \mathbb{P}_{r-1}. \end{array} \tag{1.39}$$

We define the reduced first-order $H(\text{div})$ conforming space as first-order polynomial space supplementing with the curl of second-order direct serendipity space $\mathcal{DS}_2$ as

$$\mathbb{V}_1^0(E) = (\mathbb{P}_1(E))^2 \oplus \text{curl}\mathbb{S}_2^{\mathcal{DS}}(E). \tag{1.40}$$

Here, $\mathbb{V}_1^0(E)$ can be given a global basis with the normal components of the shape functions merged continuously on the shared edge of two elements.

The dimension of this reduced first-order $H(\text{div})$ conforming space is

$$\dim(\mathbb{V}_1^0(E)) = 2 \times 3 + 2 = 8. \tag{1.41}$$

On each edge of a quadrilateral element E, two independent basis functions are defined: one with vanishing average normal flux across the edge, and the other with non-vanishing average flux.

We state the degrees of freedom of element as normal flux on each edge as

$$DOF_{e_i} = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, \lambda_1 \rangle, \quad \lambda_1 \in \mathbb{P}_1, \tag{1.42}$$

where λ_1 is a linear function restricted to edge e_i, and $\dim(\lambda_1) = 2$. We then present a construction of two independent shape functions on each edge: $\boldsymbol{\varphi}_{l,i}$ and $\boldsymbol{\varphi}_{c,i}$, such that

$$\boldsymbol{\varphi}_{c,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ 1, & j = i. \end{cases} \tag{1.43}$$

$$\boldsymbol{\varphi}_{l,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ l_i(\mathbf{x}), & j = i, \end{cases} \tag{1.44}$$

where l_i is a non vanishing linear polynomial defined on edge e_i orthogonal to one, i.e., $\int_{e_i} l_i ds = 0$.

To begin with, we take the curl of the auxiliary rational function R_i (1.17) as

$$\text{curl}R_i(\mathbf{x}) = \frac{\boldsymbol{\tau}_{i+2}(\mathbf{x})\lambda_i(\mathbf{x}) - \boldsymbol{\tau}_i(\mathbf{x})\lambda_{i+2}(\mathbf{x})}{(\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x}))^2}, \tag{1.45}$$

where $i \equiv i \pmod 4$. We first present one way to construct the "linear" shape function $\boldsymbol{\varphi}_{l,i}$ in two steps as

$$\tilde{\boldsymbol{\varphi}}_{l,i} = \mathrm{curl}(\lambda_{i+1}\lambda_{i+3}R_i) = (\boldsymbol{\tau}_{i+1}\lambda_{i+3} + \lambda_{i+1}\boldsymbol{\tau}_{i+3})R_i + \lambda_{i+1}\lambda_{i+3}\mathrm{curl}R_i. \tag{1.46}$$

We then normalize it such that l_i varying from -1 to 1 when restricted to edge e_i as

$$\boldsymbol{\varphi}_{l,i} = \frac{\tilde{\boldsymbol{\varphi}}_{l,i}(\mathbf{x})|e_i|}{4\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.47}$$

where the index $i \equiv i \pmod 4$, and the length of edge e_i is $|e_i| = |\mathbf{x}_{v,i} - \mathbf{x}_{v,i+3}|$. In Figure 1.8, we draw $\boldsymbol{\varphi}_{l,2}$ on the left element and $\boldsymbol{\varphi}_{l,0}$ on the right element sharing a edge, and show normal flux joining continuously as a linear function on the shared edge.

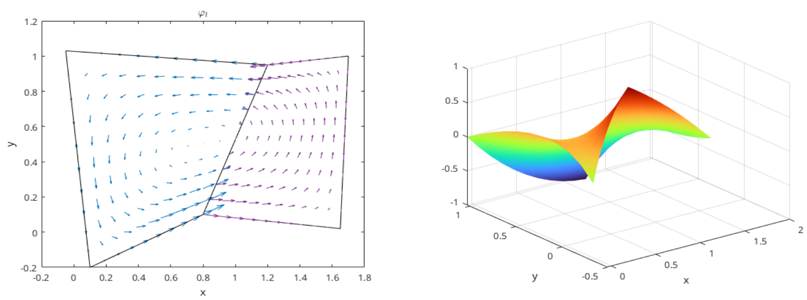


Figure 1.8: "Linear" shape function $\boldsymbol{\varphi}_l$.

The construction of a the constant basis function is a bit more complicated. We follow the process by Arbogast et al. (2022). We start by defining a linear nodal function $\phi^*_{v,i} \in \mathcal{DS}_2(E)$ such that $\phi^*_{v,i}(\mathbf{x}_{v,k}) = \delta_{ik}$. Its restriction to each edge of E is a linear function as

$$\phi^*_{v,i}(\mathbf{x}) = \varphi^2_{v,i}(\mathbf{x}) + \big[\frac{1}{2}\varphi^2_{e,i} + \frac{1}{2}\varphi^2_{e,i+1}\big], \tag{1.48}$$

where $i \equiv i \pmod 4$. We then define $\boldsymbol{\varphi}^*_{v,i}(\mathbf{x}) = \mathrm{curl}\phi^*_{v,i}$ so that

$$\boldsymbol{\varphi}^*_{v,i} \cdot \boldsymbol{\nu}_j|_{e,j} = \begin{cases} 1/|e_i|, & j = i, \\ -1/|e_i|, & j = i+1, \\ 0, & j = i+2, i+3. \end{cases} \tag{1.49}$$

The "constant" basis function on the edge can now be defined as

$$\boldsymbol{\varphi}_{c,i} = \frac{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1} |e_{i+1}| \boldsymbol{\varphi}^*_{v,i} + \mathbf{x} - \mathbf{x}_{v,i+2}}{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1} |e_{i+1}|/|e_i| + (\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_i}, \tag{1.50}$$

where $i \equiv i \pmod 4$. The "constant" shape function $\boldsymbol{\varphi}_{c,i}$ has a normal flux of a constant value 1 on edge e_i and evaluates 0 normal flux on all the other edges. In Figure 1.9, we draw $\boldsymbol{\varphi}_{c,2}$ on the left element and $\boldsymbol{\varphi}_{c,0}$ on the right element sharing a edge, and show that the normal flux joining continuously as a constant 1 on the shared edge.

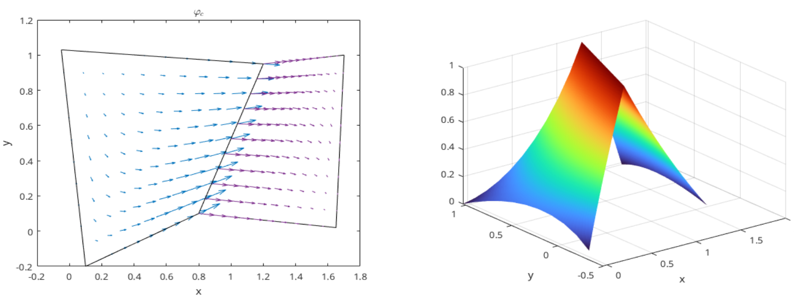


Figure 1.9: "Constant" shape function $\boldsymbol{\varphi}_c$.

1.4.3 Stability and Approximation Properties for $\mathbb{V}_1^0$

The accuracy and stability of such element has been well studied and tested in the work by Arbogast et al. (2022). We state the result directly for completeness. Consider a projection operatior π preserves element average divergence and average edge normal fluxes.

$$\pi : H(\mathrm{div}; \Omega) \cap (L^{2+\epsilon}(\Omega))^2 \to \mathbb{V}_1^0, \tag{1.51}$$

where $\epsilon > 0$. This global projection operator π is understood as combination of local operator π_E according to degrees of freedom. The operator π also satisfies the commuting diagram property as

$$\mathcal{P}_{W_0} \nabla \cdot \mathbf{v} = \nabla \cdot \pi \mathbf{v}, \tag{1.52}$$

where $\mathcal{P}_{W_0}$ is the L^2-orthogonal projection operator onto $W_0 = \nabla \cdot \mathbb{V}_1^0$, which is a space of constant values.

Lemma 1.5. *Stability and approximation properties for* $\mathbb{V}_1^0$.

Under the assumption of Lemma 1.3 and Lemma 1.2, there exists a constant $C > 0$, *independent of* $\mathcal{T}_h$ *and* $h > 0$, *such that*

$$\begin{aligned} \|\mathbf{u} - \pi_D \mathbf{u}\|_{0,\Omega} &\leq C \|\mathbf{u}\|_{k,\Omega} h^k, \qquad k = 1, \ldots, 2 \\ \|\nabla \cdot (\mathbf{u} - \pi_D \mathbf{u})\|_{0,\Omega} &\leq C \|\nabla \cdot \mathbf{u}\|_{k,\Omega} h^k, \quad k = 0, \ldots, 1 \\ \|p - \mathcal{P}_{W_s} p\|_{0,\Omega} &\leq C \|p\|_{k,\Omega} h^k, \qquad k = 0, \ldots, 1 \end{aligned} \tag{1.53}$$

Moreover, the discrete inf-sup condition holds

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^0} \frac{(\omega_h, \nabla \cdot \mathbf{v}_h)}{\|\mathbf{v}_h\|_{H(div)}} \geq \gamma \|\omega_h\|_{0,\Omega}, \quad \forall \omega_h \in W_0, \tag{1.54}$$

holds for some $\gamma > 0$ *independent of* $\mathcal{T}_h$ *and* $h > 0$.

Proof. See Arbogast et al. (2022). □

1.4.4 A Stokes Model Problem

Consider a model problem of an incompressible isotropic equation equipped with homogeneous boundary condition as

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{0} \quad \text{on } \partial\Omega. \end{aligned} \tag{1.55}$$

We again pick energy space for displacement-pressure pair $(\mathbf{u}, p)$ in identical test and trial spaces as

$$\begin{aligned} \mathbb{V}_S &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \text{ on } \partial\Omega\} = (H_0^1(\Omega))^2, \\ \mathbb{U}_S &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{0} \text{ on } \partial\Omega\} = \mathbb{V}_S, \\ \mathbb{W}_S &= \{p \in L^2(\Omega) : \int_\Omega p d\mathbf{x} = 0\} = L^2(\Omega)/\mathbb{R}, \end{aligned} \tag{1.56}$$

We write weak form with these energy space, and consider the standard variational form as:

Find $\mathbf{u} \in \mathbb{U}_S$ and $p \in \mathbb{W}_S$ such that

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) = (\mathbf{f}, \mathbf{v}), &\quad \forall \mathbf{v} \in \mathbb{V}_S, \\ (\nabla \cdot \mathbf{u}, \omega) = 0, &\quad \forall \omega \in \mathbb{W}_S. \end{aligned} \tag{1.57}$$

We introduce two finite-dimensional subspace $\mathbb{V}_{S,h} \subset \mathbb{V}$ and $\mathbb{W}_{S,h} \subset \mathbb{W}$, and consider the discretized problem:

Find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ such that

$$\begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h) - (p_h, \nabla \cdot \mathbf{v}_h) = (\mathbf{f}, \mathbf{v}_h), &\quad \forall \mathbf{v}_h \in \mathbb{V}_{S,h}, \\ (\nabla \cdot \mathbf{u}_h, \omega_h) = 0, &\quad \forall \omega_h \in \mathbb{W}_{S,h}. \end{aligned} \tag{1.58}$$

Now we construct a pair of discretized space $\mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ for this Stokes model problem in the next subsection.

1.4.5 Modified Bernardi-Raugel Element

Stable mixed finite element pairs for the Stokes problem have been widely and thoroughly studied on triangles and rectangles. One notable construction on rectangles was discussed in Bernardi and Raugel (1985) and Fortin (1981), where polynomial spaces are supplemented with bubble functions. Consider the standard type of Bernardi-Raugel (BR) element using space of tensor product polynomials

$$(\mathbb{Q}_{1,1}(E))^2 \oplus \operatorname{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} \subset (\mathbb{Q}_{2,2}(E))^2, \tag{1.59}$$

where $\mathbf{b}_i$ is a quadratic bubble function defined for each edge e_i. A similar but more general type of element was discussed by Arbogast and Wheeler (2005).

We now modify the original BR element to extend it to quadrilaterals using the direct serendipity space as

$$\mathbb{V}_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \operatorname{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} \subset (\mathcal{DS}_2(E))^2. \tag{1.60}$$

One natural choice for the supplemental bubble function $\mathbf{b}_i$ is the previously defined edge nodal basis function $\varphi_{e,i}^{(2)} \in \mathcal{DS}_2$ times the unit normal, i.e.,

$$\mathbf{b}_i(\mathbf{x}) = \varphi_{e,i}^{(2)}(\mathbf{x})\boldsymbol{\nu}_i, \tag{1.61}$$

which is quadratic when restricted to edge e_i according to (1.21). The quadratic bubble is guaranteed to be conforming since it evaluates zero on two vertices and 1 at the middle point of the edge due to normalization, which is uniquely determined by these three points. The integral of this quadratic bubble can be accurately computed by a three-point Gaussian quadrature rule.

We state the degrees of freedom of this type of element as follows.

Definition 1.5. For an element $E \in \mathcal{T}_h$, $\mathbf{u}_h$ is uniquely defined by the following degrees of freedom:

(i) for corner point v_i and Cartesian direction unit normal vectors $\{\mathbf{i}, \mathbf{j}\}$,

$$DOF_{(v_i,\mathbf{i})}^{(\mathrm{i})}(\mathbf{u}_h) = \mathbf{u}_h(v_i)\cdot\mathbf{i}, \quad DOF_{(v_i,\mathbf{j})}^{(\mathrm{i})}(\mathbf{u}_h) = \mathbf{u}_h(v_i)\cdot\mathbf{j};$$

(ii) for edge e_i,

$$DOF_{e_i}^{(ii)} = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, 1\rangle.$$

We give an explicit construction of the shape functions. We choose the nodal basis function on vertex v_i in two Cartesian directions as

$$N_{v_i,x} = \begin{bmatrix} \psi_{v,i}(\mathbf{x}) \\ 0 \end{bmatrix}, \quad N_{v_i,y} = \begin{bmatrix} 0 \\ \psi_{v,i}(\mathbf{x}) \end{bmatrix} \tag{1.62}$$

We choose the edge basis function on edge e_i as

$$N_e = \varphi_{e,i}^{(2)}\boldsymbol{\nu}_i \tag{1.63}$$

Note that bubble functions are quadratic when restricted to edge e_i, while nodal basis functions are linear when restricted to edge e_i, therefore bubble functions cannot be in the same space as nodal basis function. We conclude that this element is well-defined, i.e., the degrees of freedom uniquely determine the function.

We pair it with the space

$$\mathbb{W}_0 = \{\omega \in L^2(\Omega); \quad \omega|_E \in P_0\}/\mathbb{R}, \tag{1.64}$$

which has piece wise constant values for each element E. We provide stability and approximation analysis for this space pair $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$ in the next subsection.

1.4.6 Stability and Approximation Properties for $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$

Before analyzing the stability and approximation properties for the modified BR element, we state one useful lemma.

Lemma 1.6. *Let E be a nondengenate convex quadrilateral element that is shape regular as in definition* 1.1. *For each integer $m \geq 0$ and real number $p \in [1, \infty]$ there exist positive constants C_1 and C_2, possibly depending on σ^* but otherwise independent of the geometry of E, such that the following upper bounds hold for all $v \in W_p^m(E)$:*

$$\begin{cases} |v|_{W_p^0(E)} \leq C_1 h_E^{2/p} |\tilde{v}|_{W_p^0(\tilde{E})}, \\ |v|_{W_p^1(E)} \leq C_2(\sigma^*, p) |\tilde{v}|_{W_p^1(\tilde{E})}, \end{cases} \tag{1.65}$$

where $\tilde{E}$ represents local reference quadrilateral element $\tilde{E}$ illustrated in Figure 1.1, and $\tilde{v}$ is the function defined on the local reference element $\tilde{E}$.

Proof. See Girault and Raviart (2011). □

We reuse the interpolation operator $\mathcal{I}_h^1$ defined in equation (1.28). The projection operator $\pi \in (H_0^1(\Omega))^2 \to \mathbb{V}_1^{BR} \cap (H_0^1(\Omega))^2$ is understood as a combination of local projection operators π_E as:

(i) For each vertex $v \in \partial E$

$$\pi_E \mathbf{v}(\mathbf{x}_v) = \mathcal{I}_h^1 \mathbf{v}(\mathbf{x}_v); \tag{1.66}$$

(ii) For each edge $e \subset \partial E$,

$$\int_e (\pi_h \mathbf{v} - \mathbf{v}) \cdot \boldsymbol{\nu} ds = 0. \tag{1.67}$$

Lemma 1.7. *For all* $\mathbf{v} \in (H^k(\Omega))^2 \cap (H_0^1(\Omega))^2, \quad k = 1, 2$. *The operator* π_E *defined by* (1.66)–(1.67) *satisfies*

$$(q_h, \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v})) = 0, \quad \forall q_h \in \mathbb{W}_0 \tag{1.68}$$

Assuming shape regularity as defined in Definition 1.1, the global projection operator π *satisfies the error bound*

$$\|\mathbf{v} - \pi\mathbf{v}\|_{H^1(\Omega)} \leq Ch^m \|\mathbf{v}\|_{H^{m+1}(\Omega)}, \quad \forall \mathbf{v} \in (H^{m+1}(\Omega))^2, \; m = 0, 1. \tag{1.69}$$

Proof. From the definition of the degree of freedom (1.66) and (1.67), we decompose the projection operator π_E as

$$\pi_E \mathbf{v} = \mathcal{I}_h^1 \mathbf{v} + \sum_{i=0}^{3} \alpha_i \mathbf{b}_i, \tag{1.70}$$

where the coefficient

$$\alpha_i = \frac{\int_{e_i} (\mathbf{v} - \mathcal{I}_h^1 \mathbf{v}) \cdot \boldsymbol{\nu}_i \, ds}{\int_{e_i} \mathbf{b}_i \cdot \boldsymbol{\nu}_i \, ds}. \tag{1.71}$$

The approximation property of the interpolation operator $\mathcal{I}_h^1$ has been discussed in Lemma 1.4,

$$\left\| \mathbf{v} - \mathcal{I}_h^1 \mathbf{v} \right\|_{H^m(E)} \leq h^{2-m} |\mathbf{v}|_{H^2(E^*)}, \quad m = 0, 1, \tag{1.72}$$

where $E^* = \cup_{F \in \mathcal{T}_h, F \cap E \neq \emptyset}$ defined in Lemma 1.3.

Lemma 1.6 indicates an estimation of the bubble function as

$$|\mathbf{b}_i|_{H^m(E)} \leq C(\sigma^*) h^{1-m} |\tilde{\mathbf{b}}_i|_{H^1(\tilde{E})} \leq C(\sigma^*) h^{1-m}, \quad m = 0, 1. \tag{1.73}$$

We continue to estimate $|\alpha_i|$ as

$$\int_{e,i} \varphi_{e,i}^{(2)} ds = |e_i| \int_{\tilde{e}_i} \tilde{\varphi}_{e,i}^{(2)} \, d\tilde{s}. \tag{1.74}$$

According to Trace Theorem 1.1, a continuous trace mapping can be constructed. Combining with Lemma 1.6, we estimate the term $|\alpha_i|$ as

$$|\alpha_i| \leq C \int_{\tilde{e},i} |\tilde{\mathbf{v}} - \tilde{\mathcal{I}}_h^1 \tilde{\mathbf{v}}| \, ds \leq C h^k h^{-1} |\mathbf{v}|_{H^k(E^*)}, \quad k = 1, 2. \tag{1.75}$$

The overall estimate is then

$$|\alpha_i| |\mathbf{b}_i|_{H^m(E)} \leq C h^{2-m} |\mathbf{v}|_{H^k(E^*)}, \quad m = 0, 1. \tag{1.76}$$

Combining with the interpolation error estimate result in equation (1.72), we get a local error estimate as

$$\|\mathbf{v} - \pi_E \mathbf{v}\|_{H^1(E)} \leq C h |\mathbf{v}|_{H^2(E)}. \tag{1.77}$$

The global error estimate is obtained by combining local results together. $\square$

In addition to the approximability lemma 1.4, we also need stability result.

Lemma 1.8. *For any $\mathbf{v} \in H_0^1(\Omega)$, there exists $\mathbf{v}_h \in \mathbb{V}_1^{\mathcal{BR}}$ such that*

$$\|\mathbf{v}_h\|_{H^1(\Omega)} \leq C \|\mathbf{v}\|_{H^1(\Omega)}. \tag{1.78}$$

Proof. We express discrete solution $\mathbf{v}_h$ with the predefined projection operator $\mathbf{v}_h = \pi_h \mathbf{v}$, and denote the $\mathbf{w}_h = \mathcal{I}_h^1 \mathbf{v}$, then we have the following local inequality as

$$\|\mathbf{v}_h\| \leq \|\mathbf{w}_h\|_{H^1(E)} + h^{-1} \big\{ \|\mathbf{v} - \mathbf{w}_h\|_{L^2(E)} + h |\mathbf{v} - \mathbf{w}_h|_{H^1(E)} \big\} \leq C \|\mathbf{v}\|_{H^1(E)}. \tag{1.79}$$

The global stability result can be obtained by combining local results as well. $\square$

Theorem 1.9. *With the established approximability result stated in Lemma 1.7, and the stability result stated in Lemma 1.8, for a convex domain Ω, there exists a constant $\gamma > 0$ such that*

$$\inf_{q_h \in \mathbb{W}_0} \sup_{\mathbf{v}_h \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \mathbf{v}_h, q_h)}{\|\mathbf{v}_h\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} \geq \gamma > 0. \tag{1.80}$$

Proof.

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \mathbf{v}_h, q_h)}{\|\mathbf{v}_h\|_{H^1(\Omega)} \|\omega\|_{L^2(\Omega)}} \geq \sup_{\mathbf{v}_h \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \pi\mathbf{v}, q_h)}{\|\pi\mathbf{v}\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}}. \tag{1.81}$$

Recall that the $\inf - \sup$ condition holds for continuous space as

$$\inf_{q \in L^2(\Omega)} \sup_{\mathbf{v} \in (H^1(\Omega))^2} \frac{(\nabla \cdot \mathbf{v}, q)}{\|\mathbf{v}\|_{H^1(\Omega)} \|q\|_{L^2(\Omega)}} \geq \gamma > 0, \tag{1.82}$$

along with the condition $(q_h, \nabla \cdot (\mathbf{v} - \pi_E \mathbf{v})) = 0$ proved by Lemma (1.7) we have

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \pi\mathbf{v}, q_h)}{\|\pi\mathbf{v}\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} = \sup_{\mathbf{v}_h \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \mathbf{v}, q_h)}{\|\pi\mathbf{v}\|_{H^1(\Omega)} \|q_h\|_{L^2(\Omega)}} \geq \frac{\gamma}{C} > 0. \tag{1.83}$$

□

We conclude that our space $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$ forms a inf-sup stable pair with first order overall accuracy.

1.4.7 Numerical Test

We test our modified Bernardi-Raugel elements on the incompressible isotropic elasticity problem but with a non-homogenous Dirichlet boundary condition, i.e., on the problem:

Find the displacement $\mathbf{u}$ and pressure p such that

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{g} \quad \text{on } \partial\Omega. \end{aligned} \tag{1.84}$$

We define a simple choice of energy space

$$\begin{aligned} \mathbb{V} &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \quad \text{on } \partial\Omega\}, \\ \mathbb{U} &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{g} \quad \text{on } \partial\Omega\} = \tilde{\mathbf{g}} + \mathbb{V}, \\ \mathbb{W} &= \{p \in L^2(\Omega)/\mathbb{R}\}, \end{aligned} \tag{1.85}$$

where $\tilde{\mathbf{g}}$ is a finite energy lift. We assume that $\mathbf{g} \in (H^{1/2}(\partial\Omega))^d$ and understand $\tilde{\mathbf{g}} \in (H^1(\Omega))^d$ as a continuous extension from the boundary into the domain. We write the problem in weak form as:

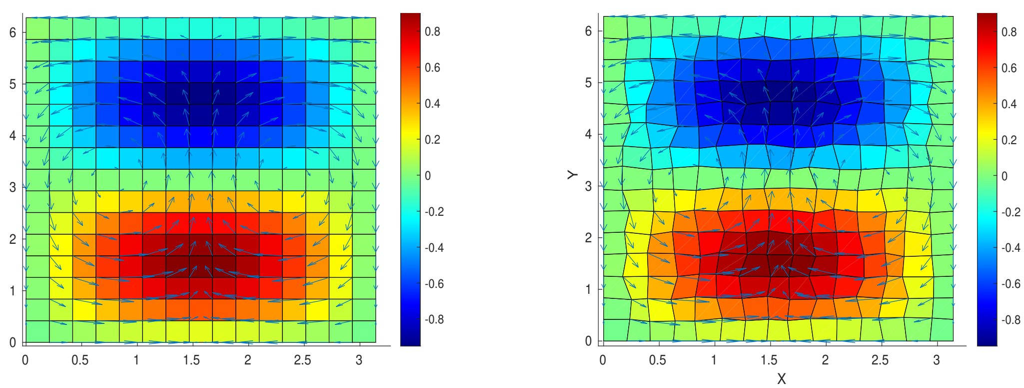
Find $\mathbf{u} \in \mathbb{U}$ and $p \in L^2(\Omega)$ such that

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{V}, \\ (\nabla \cdot \mathbf{u}, \omega) &= 0, \qquad\quad \forall \omega \in \mathbb{W}. \end{aligned} \tag{1.86}$$

We consider the true solution of test problem (1.86) defined on a square region $\Omega = [0, \pi] \times [0, 2\pi]$. We manufacture a displacement and pressure as

$$\mathbf{u} = \begin{bmatrix} \cos(x)\sin(y) \\ -\sin(x)\cos(y) \end{bmatrix}, \quad p = \sin(x)\sin(y), \tag{1.87}$$

which has 0 normal flux on the boundary. We deal with the constant kernal by enforcing a condition that $\int_\Omega p \, d\mathbf{x} = 0$ and solve the linear system directly. The numerical results are computed on rectangular and logically rectangular quadrilateral meshes as depicted in Figures 1.10a and 1.10b, respectively. We calculate the L^2 norm of the residual (i.e., the difference) of the pressure, velocity, and derivative of velocity.



(a) Rectangular Grids. Shown are pressure (color) and velocity (arrows).

(b) Distorted Grids. Shown are pressure (color) and velocity (arrows).

Figure 1.10: Numerical results on rectangular and quadrilateral mesh.

In Table 1.1 and Table 1.2 we display the result of a calculation of convergence on $n \times n$ rectangular and distorted grids, respectively, for $n = 8, 16, 32, 64$. A second order convergence rate is achieved by velocity and pressure due to mid point integration rule. A first order convergence rate is achieved by derivative of velocity. Both meet our expectations for optimal convergence order.

Grid size	8×8	16×16	32×32	64×64
$L^2(\text{res}_u)$	4.502E−1	1.052E−1	2.513E−2	6.130E−3
$L^2(\text{res}_{du})$	1.039E+0	4.463E−1	2.093E−1	1.019E−1
$L^2(\text{res}_p)$	2.632E+0	5.845E−1	1.374E−1	3.330E−2

Table 1.1: Rectangular Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

Grid size	8×8	16×16	32×32	64×64
$L^2(\text{res}_u)$	4.603E−1	1.074E−2	2.567E−2	6.266E−3
$L^2(\text{res}_{du})$	1.070E+0	4.670E−1	2.207E−1	1.078E−1
$L^2(\text{res}_p)$	2.685E+0	5.981E−1	1.525E−1	4.580E−2

Table 1.2: Distorted Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

1.5 Coupled Degenerate Darcy-Stokes System

Now we consider the nondimensionalized scaled degenerate Darcy-Stokes coupled boundary-value problem:

Find two velocity-pressure potential pairs $(\check{\mathbf{v}}_r, \check{q}_l)$ and $(\check{\mathbf{v}}_s, \check{q})$ such that

$$\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) = 0, \tag{1.88}$$

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.89}$$

$$\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi)\mathbf{n}, \tag{1.90}$$

$$\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.91}$$

and

$$\begin{aligned} \check{\mathbf{v}}_r &= \mathbf{g}_r \quad \text{on } \Gamma_{u,D},\\ \check{\mathbf{v}}_s &= \mathbf{g}_s \quad \text{on } \Gamma_{u,S},\\ p &= p_0 \quad \text{on } \Gamma_{i,D},\\ \boldsymbol{\tau}\cdot\boldsymbol{\nu} - p &= \boldsymbol{t}_0 \quad \text{on } \Gamma_{i,S}, \end{aligned} \tag{1.92}$$

where $\Gamma_{u,D}$ and $\Gamma_{u,S}$ denote the part of boundary we assign essential boundary conditions for Darcy and Stokes problem respectively, and $\Gamma_{i,D}$ and $\Gamma_{i,S}$ represent the part of boundary we assign natural boundary conditions respectively. We have $\Gamma_{u,D} \cup \Gamma_{i,D} = \partial\Omega$ and $\Gamma_{u,S} \cup \Gamma_{i,S} = \partial\Omega$.

We follow the procedure by Arbogast et al. (2017) to discretize this boundary value problem. To begin with, a scaled Sobolev space is defined as follows for this specific scaled formulation, where the scaled velocity $\check{\mathbf{v}}_r$ should lie in.

Definition 1.6. Consider a scaled space:

$$\mathbb{V}_D^\phi = H_\phi(\mathrm{div};\Omega) = \Big\{ \mathbf{v} \in (L^2(\Omega))^2 : \phi^{-1/2}\nabla \cdot (\phi^{1+\Phi}\mathbf{v}) \in L^2(\Omega) \Big\}, \tag{1.93}$$

which is a Hilbert space equipped with the inner product

$$(\mathbf{u},\mathbf{v})_{\mathbb{V}_D^\phi} = (\mathbf{u},\mathbf{v}) + (\phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\mathbf{u}), \phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\mathbf{v})). \tag{1.94}$$

The normal trace on $\partial\Omega$ is well-defined as

$$\left\| \phi^{1/2+\Theta}\mathbf{v}_\phi \cdot \nu \right\|_{-1/2,\partial\Omega} < C_\Omega \Big\{ \left\| \mathbf{v}_\phi \right\| + \left\| \phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\mathbf{v}_\phi) \right\| \Big\} \tag{1.95}$$

subsequently, we have the space $H_\phi^{-1/2}(\partial\Omega)$ understood as the image of this normal trace operator.

1.5.1 The weak formulation

We define the following function spaces

$$\begin{aligned} \mathbb{V}_D^\phi &= \Big\{ \mathbf{v} \in H_\phi(\mathrm{div};\Omega) : \phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu} = 0 \text{ on } \Gamma_{u,D} \Big\}, \\ \mathbb{W}_D &= L^2(\Omega), \\ \mathbb{V}_S &= \Big\{ \mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{u,S} \Big\}, \\ \mathbb{W}_S &= L^2(\Omega), \end{aligned} \tag{1.96}$$

where each space is equipped with their natural norm. To empose essential boundary conditoins, we assume that a function $\mathbf{g}_s = \check{\mathbf{v}}_{s,0}$ on $\Gamma_{u,S}$ and $\mathbf{g}_s = \mathbf{0}$ on $\partial\Omega/\Gamma_{u,S}$. Then $\mathbf{g}_s \in (H^{1/2}(\partial\Omega))^2$ and extended it continuously from the boundary to the domain, so that the extension $\mathbf{g}_s \in (H^1(\Gamma_{u,S}))^2$ and $\|\mathbf{g}_{s,0}\|_1 \le C\,\|\mathbf{v}_{s,0}\|_{1/2,\Gamma_{u,S}}$. Similarly, we can define a bounded extension of $\check{\mathbf{v}}_r \in \mathbb{V}_D^\phi$ such that

Scaled nondimensionalized weak form. Find $\check{\mathbf{v}}_r \in \mathbb{V}_D^{\phi} + \tilde{\mathbf{g}}_r$, $\check{q}_l \in \mathbb{W}_D$, $\check{\mathbf{v}}_s \in \mathbb{V}_S + \tilde{\mathbf{g}}_S$, and $\check{q} \in \mathbb{W}_S$ such that

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_s), \nabla\boldsymbol{\psi}_s) - (\check{q}, \nabla\cdot\boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s) \qquad \forall\boldsymbol{\psi}_s \in \mathbb{V}_S, \tag{1.97}$$

$$(\nabla\cdot\check{\mathbf{v}}_s, \omega) + \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega\Big) = 0 \qquad \forall\omega \in \mathbb{W}_S, \tag{1.98}$$

$$(\check{\mathbf{v}}_r, \boldsymbol{\psi}_r) - (\check{q}_l, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r)) = 0, \qquad \forall\boldsymbol{\psi}_r \in \mathbb{V}_D^{\phi}, \tag{1.99}$$

$$(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r), \omega_f) + \Big(\frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega_f\Big) = 0 \qquad \forall\omega_f \in \mathbb{W}_D. \tag{1.100}$$

The Existence and uniquense of the solution has been proved by Arbogast et al. (2017). We will not repeat the proof here.

1.5.2 MFEM discretization

We discretize the formed degenerate system with previously defined $H(\mathrm{div})$ and H^1 conforming space. We write the function spaces out as

$$\begin{aligned}\mathbb{V}_1^0 &= (\mathbb{P}_1)^2 \oplus \mathrm{curl}\mathbb{S}_2^{\mathcal{DS}}\\ \mathbb{W} &= \mathbb{P}_0\\ \mathbb{V}_1^{\mathcal{BR}} &= \mathcal{DS}_1 \oplus \mathrm{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}\\ \mathbb{W} &= \mathbb{P}_0.\end{aligned} \tag{1.101}$$

We can impose essential boundary conditions

Scaled mixed finite element method. Find $\check{\mathbf{v}}_{r,h} \in \mathbb{V}_1^0 + \hat{\mathbf{g}}_r$, $\check{q})_{r,h} \in \mathbb{W}_D$, $\mathbf{v}_{s,h} \in \mathbf{V}_S + \hat{\mathbf{g}}_s$ and $\check{q}_h \in \mathbb{W}_S$ such that

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_{s,h}), \nabla\boldsymbol{\psi}_s) - (\check{q}_h, \nabla\cdot\boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s) \qquad \forall\boldsymbol{\psi}_s \in \mathbb{V}_1^{\mathcal{BR}}, \tag{1.102}$$

$$(\nabla\cdot\check{\mathbf{v}}_{s,h}, \omega) - \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega\Big) = 0 \qquad \forall\omega \in \mathbb{W}_0^{\mathcal{BR}}, \tag{1.103}$$

$$(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_r) - (\check{q}_{l,h}, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})) = 0, \qquad \forall\boldsymbol{\psi}_r \in \mathbb{V}_1^0, \tag{1.104}$$

$$(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}), \omega_f) + \Big(\frac{1}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega_f\Big) = 0 \qquad \forall\omega_f \in \mathbb{W}_0. \tag{1.105}$$

1.5.3 Local Mass Conservation

Before we continue the discussion, we need to make sure the formulation is well-defined by not dividing by zero. The term $\phi^{-1/2}$ in the symmetric equations (1.104) and (1.105) causes trouble when there is no melting, i.e., $\phi = 0$. In order to deal with it, we first define a element-averaged $\overline{\phi}_E$ as

$$\overline{\phi}_E = \frac{1}{|E|}\int_E \phi(\mathbf{x})d\mathbf{x}, \tag{1.106}$$

where $|E|$ is the area of element E. Let

$$\phi_E = \begin{cases} \overline{\phi}_E & \text{if } \overline{\phi}_E \neq 0, \\ 0 & \text{if } \overline{\phi}_E = 0. \end{cases} \tag{1.107}$$

We now replace the $\phi^{-1/2}$ with $\phi_E^{-1/2}$. Then we expand the integral with divergence theorem as

$$\begin{aligned} \int_E \phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})\omega_l = &\int_{\partial E}\phi_E^{-1/2}\phi^{1+\Theta}\boldsymbol{\psi}_{r,h}\cdot\boldsymbol{\nu}\omega_f ds \\ &- \int_E \phi_E^{-1/2}\phi^{1+\Theta}\boldsymbol{\psi}_{r,h}\cdot\nabla\omega_l d\mathbf{x}. \end{aligned} \tag{1.108}$$

The second term vanishes when ω_l is constant on element E. We define an auxiliary auxiliary space ω_E begin 1 on E and 0 elsewhere. We sum equations (1.103) and (1.105) together assuming that $\omega = \omega_E \in \mathbb{W}_S$ and $\omega_l = \phi_E^{-1/2}\omega_E \in \mathbb{W}_D$. We now propose our local mass conserved version by replacing the term $\phi^{1/2}$ in equation (1.103) with $\phi_E^{1/2}$ as

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_{s,h}),\nabla\boldsymbol{\psi}_s) - (\check{q}_h,\nabla\cdot\boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n},\boldsymbol{\psi}_s) \qquad \forall\boldsymbol{\psi}_s\in\mathbb{V}_1^{\mathcal{BR}}, \tag{1.109}$$

$$(\nabla\cdot\check{\mathbf{v}}_{s,h},\omega) - \left(\frac{\phi\phi_E^{-1/2}}{1-\phi}(\check{q}_{l,h} - \phi_E^{1/2}\check{q}_h),\omega\right) = 0 \qquad \forall\omega\in\mathbb{W}_0^{\mathcal{BR}}, \tag{1.110}$$

$$(\check{\mathbf{v}}_{r,h},\boldsymbol{\psi}_r) - (\check{q}_{l,h},\phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})) = 0, \qquad \forall\boldsymbol{\psi}_r\in\mathbb{V}_1^0, \tag{1.111}$$

$$(\phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}),\omega_f) + \left(\frac{\phi\phi_E^{-1}}{1-\phi}(\check{q}_{l,h} - \phi_E^{1/2}\check{q}_h),\omega_f\right) = 0 \qquad \forall\omega_f\in\mathbb{W}_0. \tag{1.112}$$

To see local mass conservation of the fluid, let $E \in \mathcal{T}_h$ be any element. The equation (1.112) gives

$$\int_E \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_{r,h}) d\mathbf{x} + \int_E \frac{\phi}{1-\phi} (\check{q}_{l,h} - q_h) dx = 0, \tag{1.113}$$

and the equation (1.110) gives

$$\int_E \nabla \cdot \mathbf{v}_{s,h} d\mathbf{x} - \int_E \frac{\phi}{1-\phi} (q_{l,h} - q_h) dx = 0. \tag{1.114}$$

By summing the two individual equations above, we can achieve local mass conservation of the phase averaged mixture as

$$\int_E \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_r + \check{\mathbf{v}}_s) d\mathbf{x} = 0. \tag{1.115}$$

1.5.4 Convergence Rate

The existence of a unique solution can be shown

We derive a bound for the error by taking the difference of

We use the projection operators defined in

1.6 Saddle Point System

Saddle point system is widely spread in technical and science scenarios, such as constrained optimization problem, image reconstruction and optimal control. Saddle point system has been widely and systematically studied, e.g., by Benzi et al. (2005). In our case, saddle point system is formed along with the Mixed Finite Element discretization. In this section, we give a brief overview of properties of a saddle point system and some common ways to solve such system. An abstract saddle point system can be represented as

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{g} \end{bmatrix}. \tag{1.116}$$

Assuming the following properties:

- $\mathbf{A}$ is symmetric: $\mathbf{A} = \mathbf{A}^T$;
- $\mathbf{A}$ is nonsingular: $\mathbf{A}^{-1}$ exists;
- $\mathbf{C}$ is symmetric and positive semidefinite;
- $\mathbf{B}$ is rank deficient.

1.6.1 Reduced Linear System

Now we discuss the saddle point system (SPS) induced by MFEM method in details including the boundary conditions. To begin with, we regroup dofs according to the boundary condition.

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_K & \mathbf{A}_M \\ \mathbf{A}_M^T & \mathbf{M} \end{bmatrix}, \tag{1.117}$$

where $\mathbf{A}_K$ consists of interaction within interior dofs and natural boundary condition dofs, and $\mathbf{A}_M$ consists of iteraction between interior dofs, natural boundary dofs and essential boundary dofs.

We perform the same treatment for matrix $\mathbf{B}$ and right hand side vector $\mathbf{f}$, by regrouping dofs according to its iteration with boundary conditions.

$$\mathbf{B} = \begin{bmatrix} \mathbf{B}_K \\ \mathbf{B}_M \end{bmatrix}, \text{and} \quad \mathbf{F} = \begin{bmatrix} \mathbf{F}_K \\ \mathbf{F}_M \end{bmatrix} \tag{1.118}$$

We rewrite the linear system with the regrouped dofs and form a reduced system as

$$\begin{bmatrix} \mathbf{A}_K & \mathbf{B}_K \\ \mathbf{B}_K^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_K - \mathbf{A}_M \mathbf{g} - \mathbf{g}_N \\ \mathbf{G} - \mathbf{B}_M^T \mathbf{g} \end{bmatrix}, \tag{1.119}$$

where $\mathbf{g}$ denotes the values prescribed for essential boundary, and $\mathbf{g}_N$ is the vector generated by the natural boundary condition part. The reformed reduced system is still a saddle point syste.

Now we take a closer look at the formed linear system. The no melting (zero porosity $\phi = 0$) region adds difficulties to solving the formed saddle point system.

On one hand, the existence of zero porosity region guarantees that $\mathbf{B}$ matrix formed in Darcy system does not has full column rank. Therefore the schur complement

$$\mathbf{S} = -(\mathbf{C} + \mathbf{B}\mathbf{A}^{-1}\mathbf{B}^T),$$

is usually not invertable directly. On the other hand, in addition to the constant null space induced by gradient of pressure, the null space of $\mathbf{B}$ matrix $\ker(\mathbf{B})$ in our darcy problem is not fixed due to evloution of melting (change of zero porosity region).

Taking into consideration that the solver will be implemented in parallel, a iterative solver is more suitable than direct solver.

1.6.2 Schur Complement

To begin with, we discuss a "direct" way to solve this derived saddle point system. Now consider a general saddle point system (1.116) with symmetric positive-definite matrix $\mathbf{A}$ as

$$\begin{aligned} \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} &= \mathbf{f}, \\ \mathbf{B}^T\mathbf{x} - \mathbf{C}\mathbf{y} &= \mathbf{g}. \end{aligned} \tag{1.120}$$

We represent the solution vector $\mathbf{x}$ as

$$\mathbf{x} = \mathbf{A}^{-1}(\mathbf{f} - \mathbf{B}\mathbf{y}), \tag{1.121}$$

where $\mathbf{A}^{-1}$ is computed with conjugate gradient algorithm. We then eliminate $\mathbf{x}$ to represent $\mathbf{y}$ as

$$\mathbf{y} = \mathbf{S}^{-1}(\mathbf{g} - \mathbf{B}^T\mathbf{A}^{-1}\mathbf{f}), \tag{1.122}$$

where $\mathbf{S}^{-1}$ is computed with Minimal Resigual algorithm Paige and Saunders (1975) due to the formed Schur complement being a symmtric indefinite system.

1.6.3 Uzawa Iteration

Uzawa algorithm was originally introduced by K.J.Arrow et al. (1958), which gained popularity in computational fluid dynamics. It has later been applied to the case of discretizing Stokes problem with Mixed Finite Element. Some inexact inner solver to accelerate uzawa iteration was discussed by Elman and Golub (1994) and Bramble et al. (1997).

We start by stating an abstract inexact ezawa iteration algorithm in Algorithm 1. Some choose diagonal matrix $\alpha\mathbf{I}$ for precondictioner $\mathbf{Q}_A$. However, such

Algorithm 1 Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$, tolence tol $= 1$ and residual $res = 0$.
2: **while** res $>$ tol **do**
3: Obtain $\tilde{\mathbf{x}} = \mathbf{Q}_A^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
4: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \tilde{\mathbf{x}}$
5: Obtain $\tilde{\mathbf{y}} = \mathbf{Q}_B^{-1}(\mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G})$
6: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$
7: Update res $= \|\tilde{\mathbf{x}}\|_2 + \|\tilde{\mathbf{y}}\|_2$
8: **end while**

simple matrix wouldn't generate a convergence result when applied to our coupled problem. In the next section, we introduce a modified algorithm that would solve our system.

1.6.4 Modified Uzawa Iteration

The linear system formed from the equations (1.109), (1.110), (1.111) and (1.112) is

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}, \tag{1.123}$$

wherein the solution represents the degrees of freem of primary variables with respect to the bases of the finite element spaces. (Insert details of calculation of each blocked matrix). We form a double saddle point system by symmetrically permutate original linear system,

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \\ \mathbf{g}_s \\ \mathbf{g}_d \end{bmatrix}. \tag{1.124}$$

We group these block matrix to form a concise version,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}, \tag{1.125}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{1.126}$$

where $\mathbf{A}$ is a positive definite matrix. However, $\mathbf{B}$ is usually a singular matrix when no melting region presents. We will present a modified Uzawa iteration algorithm to deal with this coupled degenerate singular system.

Algorithm 2 Modified Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$, tolence tol $= 1$ and residual $res = 0$.

2: **while** res $>$ tol **do**

3: Obtain $\tilde{\mathbf{x}} = \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$

4: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \tilde{\mathbf{x}}$

5: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$

6: Obtain $\tilde{\mathbf{y}} = \mathrm{argmin}_{\mathbf{x}\in V}||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$

7: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

8: Update res $= \|\tilde{\mathbf{x}}\|_2 + \|\tilde{\mathbf{y}}\|_2$

9: **end while**

1.6.5 Numerical Test

We test our MFEM setup with three sets of problems. First, we test with a manufactured solution all prescribed with essential boundary condition, but the right hand side was modified. Second, we test a pseudo 1D problem with 2D elements with three different porosity scenarios. Third, we test a more complex yet a more realistic problem representing Mid Ocean Ridge.

Problem Setup 1: Manufactured Solution.

We solve a manufactured system of equations on the rectangular domain $(x, y) \in [-1, 1]^2$. We assume a constant porosity acorss the domain as ϕ. We compute the right hand side with prescribed functions as

$$\check{q} = 0 \qquad \check{q}_f = 2(x+y) \qquad \check{\mathbf{v}}_r = -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \qquad \check{\mathbf{v}}_s = \frac{\phi_f^{1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \tag{1.127}$$

$$\nabla \cdot \check{\mathbf{v}}_r = \frac{-2\phi_f^{-1/2}}{1-\phi_f}(x+y) \tag{1.128}$$

$$\mathcal{D}\check{\mathbf{v}}_s = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \qquad (\nabla \cdot \check{\mathbf{v}}_s)\mathrm{I} = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x+y & 0 \\ 0 & x+y \end{bmatrix} \tag{1.129}$$

Substitute the compuated results into PDE

$$\begin{aligned} \tilde{\mathbf{v}}_r + \phi_f^{1/2}\nabla\check{q}_f &= -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + \phi_f^{1/2}\begin{pmatrix} 2 \\ 2 \end{pmatrix} \\ \phi_f^{1/2}\nabla \cdot (\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}\check{q}_f &= \frac{-2}{1-\phi_f}(x+y) + \frac{1}{1-\phi_f}2(x+y) = 0 \\ \nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}\check{q}_f &= \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) - \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) = 0 \\ -2(1-\phi_f)\nabla \cdot \sigma(\check{\mathbf{v}}_s) &= -2(1-\phi_f)\nabla \cdot (\mathcal{D}\check{\mathbf{v}}_s - \frac{1}{3}(\nabla \cdot \check{\mathbf{v}}_s)\mathrm{I}) = -\frac{8\phi_f^{1/2}}{3}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{aligned} \tag{1.130}$$

Problem Setup 2: Column Compaction.

In this problemsetup, we perform convergence test on a 2D grid with a 1D problem. We study a compacting column in one direction. The column extends over $y \in [-L, L]$ and has no flow through top and bottom boundaries. We prescribe the following boundary conditions

- Bottom Edge;
  - Stokes velocity $\check{\mathbf{v}}_s = \mathbf{0}$;
  - Darcy velocity $\check{\mathbf{v}}_l = \mathbf{0}$;
  - Normal flux is also determined to be zero.
- Left and Right side Edges;
  - y component of Stokes velocity is free, x component is kept as zero;
  - Darcy velocity, which is understood as flux is also kept zero;
  - normal flux dof fixed with a zero flux.
- Top Edge;
  - Stokes velocity $\check{\mathbf{v}}_s = \mathbf{0}$;
  - Darcy velocity $\check{\mathbf{v}}_l = \mathbf{0}$;
  - Normal flux is also determined to be zero.

We assume no velocity in x direction, $\check{\mathbf{v}}_s \cdot \mathbf{i} = 0$, and $\check{\mathbf{v}}_r \cdot \mathbf{i} = 0$, where $\mathbf{i}$ is the unit vector in x direction pointing from left to right. Let $u = \phi \check{\mathbf{v}}_r \cdot \mathbf{j}$ and $v = \check{\mathbf{v}}_s \cdot \mathbf{j}$, where $\mathbf{j}$ is the unit vector in y direction pointing upwards.

$$\begin{aligned} u + \phi^2 q_l' &= 0, \\ u' + \phi(q_l - q_s) &= 0, \\ [q_s - \frac{1}{3}(1-4\phi)v']' &= 0, \\ v' - \phi(q_f - q_s) &= 0, \end{aligned} \tag{1.131}$$

When no melting presents $\phi = 0$, the equations reduce to

$$u = -v = 0, \quad \text{and} \quad q_s' = 1, \quad \text{and} \quad q_l = 0. \tag{1.132}$$

When melting presents $\phi > 0$, the equations can be reduced to

$$\phi^{2+2\Theta}\left(\frac{3+\phi-4\phi^2}{3\phi}u'\right)' - u = \phi^{2+2\Theta}(1-\phi), \tag{1.133}$$

where all other quantities can be determined according to u as

$$v = -u, \quad \text{and} \quad q_l' = -\phi^{-2-2\Theta} u, \quad \text{and} \quad qs = u'/\phi + q_l. \tag{1.134}$$

When ϕ is constant, we define an auxiliary function R as

$$R = \left(\phi^{2+2\Theta} \frac{3 + \phi - 4\phi^2}{3\phi} \right)^{-1/2}. \tag{1.135}$$

We test our MFEM code on three porosity settings, and use the closed form solutions derived in Arbogast et al. (2017).

- Constant porosity;

$$\phi_0(z) = \phi_0; \tag{1.136}$$

Closed form solution:

$$\begin{aligned}
u &= -\phi_0^2(1-\phi_0)\left[1 - \frac{\cosh(Rz)}{\cosh(RL)}\right], \\
v &= -u, \\
q_l &= (1-\phi_0)\left[z - \frac{1}{R}\frac{\sinh(Rz)}{\cosh(RL)}\right] + c, \\
q_s &= (1-\phi_0)\left[z - \frac{1-4\phi_0}{3+\phi_0-4\phi_0^2}\frac{\phi_0}{R}\frac{\sinh(Rz)}{\cosh(RL)}\right] + c,
\end{aligned} \tag{1.137}$$

where c is a constant kernel.

	u		v		q_l		q_s	
$m \times n$	L^2 error	rate	L^2 error	rate	L^2 error	rate	L^2 error	rate
2×20	3.100e-05	-	3.100e-05	-	fill	-	fill	-
2×40	8.047e-06	1.95	8.047e-06	1.95	fill	-	fill	-
2×80	2.031e-06	1.99	2.031e-06	1.99	fill	-	fill	-
2×160	5.091e-07	2.00	5.091e-07	2.00	fill	-	fill	-

Table 1.3: Constant porosity compaction column test.

- Piece wise constant porosity;

$$\phi_J(z) = \begin{cases} 0 & \text{if } z > 0, \\ \phi_0 & \text{if } z \le 0, \end{cases} \tag{1.138}$$

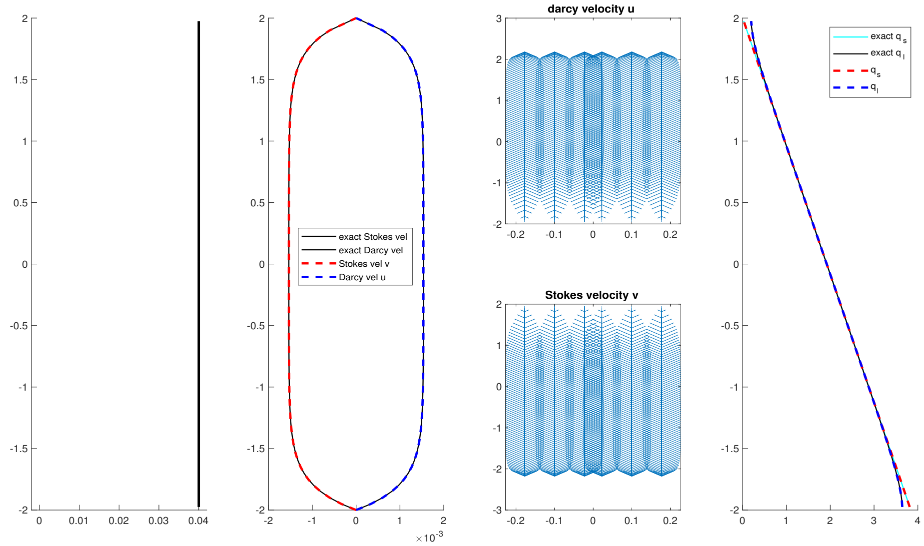


Figure 1.11: Constant porosity compaction column test.

	u		v		q_l		q_s	
$m \times n$	L^2 error	rate	L^2 error	rate	L^2 error	rate	L^2 error	rate
2×20	2.980e-05	-	2.980e-05	-	fill	-	fill	-
2×40	7.681e-06	1.96	7.681e-06	1.96	fill	-	fill	-
2×80	1.958e-06	1.97	1.958e-06	1.97	fill	-	fill	-
2×160	4.974e-07	1.98	4.974e-07	1.98	fill	-	fill	-

Table 1.4: Constant porosity compaction column test.

Closed form solution:

We conclude that $u = -v = 0$ when $z > 0$.

$$\begin{aligned} u &= \phi_0^2(1-\phi_0)\left[1 - \cosh(Rz) + \frac{1-\cosh(RL)}{\sinh(RL)}\sinh(Rz)\right] \\ v &= -u \\ q_s &= z - \frac{1-\cosh(RL)}{\sinh(R)} \end{aligned} \tag{1.139}$$

- Quadratic porosity.

$$\phi_2(z) = \begin{cases} 0 & \text{if } z > 0, \\ \phi_0 z^2 & \text{if } z \le 0, \end{cases} \tag{1.140}$$

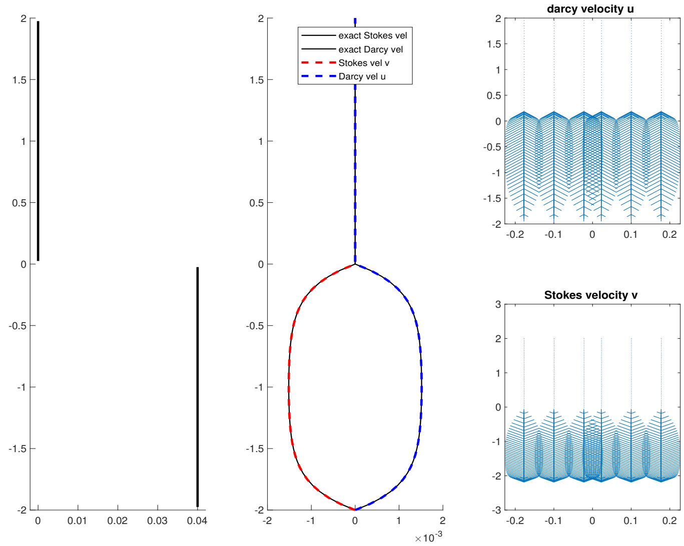


Figure 1.12: Piecewise constant porosity compaction column test.

Closed form solution:

We conclude that $u = -v = 0$ when $z > 0$.

$$\begin{aligned} u &= \frac{\phi_0^2}{1-4\phi_0}\Big(L^{4-r_1}z^{r_1} - z^4\Big), \\ v &= -v \\ q_l &= \frac{1}{1-4\phi_0}\left(z + \frac{L^{4-r_1}(-z)^{r_1-3}}{r_1-3}\right) + c, \\ q_s &= z + c, \end{aligned} \tag{1.141}$$

where

$$r_1 = \frac{3+\sqrt{9+4/\phi_0}}{2}. \tag{1.142}$$

Problem Setup 3: Mid-Ocean Ridge.

We now redo the Mid-Ocean ridge test case in Arbogast et al. (2017). We solve the full system of equations on the rectangular domain $-160\text{km} < x < 160\text{km}$, $0, x, 160\text{km}$. The boundary conditions are imposed by the constant porosity corner

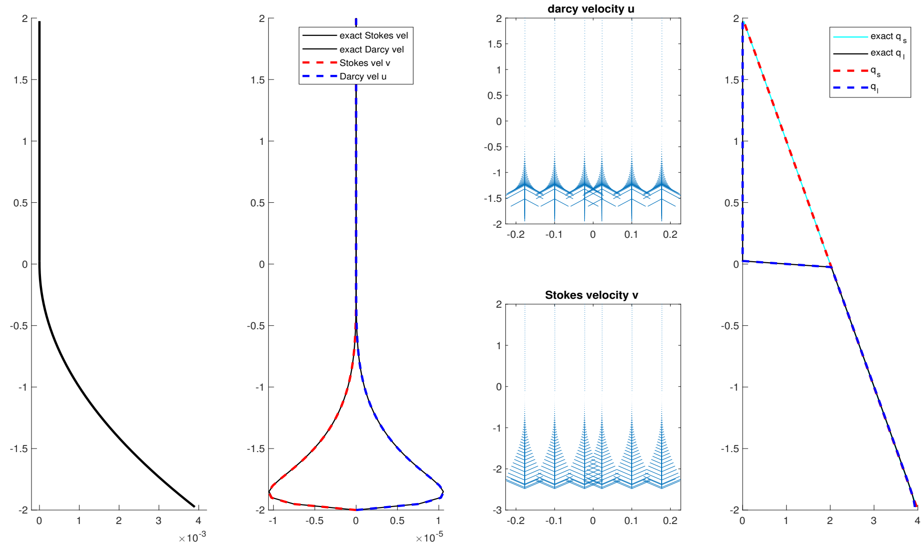


Figure 1.13: Quadratic porosity compaction column test.

	u		v		q_l		q_s	
$m \times n$	L^2 error	rate	L^2 error	rate	L^2 error	rate	L^2 error	rate
2×20	9.616e-07	-	9.616e-07	-	fill	-	fill	-
2×40	3.212e-07	1.58	3.212e-07	1.58	fill	-	fill	-
2×80	9.028e-08	1.83	9.028e-08	1.83	fill	-	fill	-
2×160	2.746e-08	1.72	2.746e-08	1.72	fill	-	fill	-

Table 1.5: Constant porosity compaction column test.

flow solution of Stokes velocity as

$$\mathbf{v}_s = \frac{2U_0}{\pi(x^2+z^2)} \begin{pmatrix} \tan^{-1}(x/z)(x^2+z^2) - xz \\ -z^2 \end{pmatrix}, \tag{1.143}$$

where $U_0 = 10^{-9}$m/s $= 3.2$cm/yr is the constant upwelling velocity of mantle. According to mass conservation condition, we set no Darcy flux across the boundary. We then set a no identically vanishing porosity by the formula

$$\phi(x,z) = \begin{cases} 0.05(\frac{120\text{km} - z}{120\text{km}})^2(1 - \frac{|x|}{z+l}), & \text{if } x \leq 120\text{km and } |x| \leq z + l, \\ 0, & \text{otherwise,} \end{cases} \tag{1.144}$$

where $l = 20$m is the offset set to avoid singularity computation. at the center $x = 0$ we translate the coordinates x to $x - l$ if $x < 0$, and $x + l$ when $x > 0$.

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